Self-Supervised Spatiotemporal Imputation Model for Highly Sparse Chl-a Data via Fusing Multisource Satellite Data

Shuyu Wang, Wengen Li, *Member, IEEE*, Jihong Guan, Xiwei Liu, *Senior Member, IEEE*, Yichao Zhang, *Member, IEEE*, and Shuigeng Zhou, *Senior Member, IEEE*

***Abstract*— Monitoring Chlorophyll-a (Chl-a) concentration in ocean is of considerable significance for early warning of algae disasters, marine ecological environment protection, etc. However, due to various uncontrollable factors such as cloud cover and thick aerosols, the missing rate of observed Chl-a data is quite high, which seriously hinders its applications. The existing data imputation methods are mostly applicable to Chl-a data with a low missing rate, and perform much worse when the missing rate reaches 0.7 or above. To address this issue, we proposed a self-supervised spatiotemporal imputation (S2-STI) model for highly sparse Chl-a data imputation by fusing multisource satellite data. First, to obtain comprehensive information about the missed Chl-a data, we designed a multisource sparse data fusion (MSDF) module to fuse multisource satellite data, including Chl-a data, sea surface temperature (SST) data, and photosynthetically available radiation data. MSDF constructs a spatialtemporal graph network to learn high-quality spatiotemporal representations of SST and photosynthetically active radiation (PAR) data, and fuses the learned representations with Chl-a data to enrich the information for imputation. Then, we developed a generation-based data imputation (GDI) module to model the distribution of Chl-a data based on the outputs of MSDF. Considering the high data sparsity, we designed a self-supervised training strategy to train S2-STI in the absence of ground truth. Finally, we leverage the data generated by the GDI module in the trained S2-STI model to fill in missing values in the sparse Chl-a data. Experiments on real datasets show that S2-STI achieves much better performance than the existing data imputation methods. Specifically, for the Chl-a data with a missing rate of 0.9, S2-STI improves by at least 12% in terms of masked autoencoder (MAE) error when compared with the strong baseline methods.**

***Index Terms*— Chlorophyll-a (Chl-a) concentration, data imputation, high sparsity, multisource satellite data, spatiotemporal modeling.**

I. Introduction

THE Chlorophyll-a (Chl-a) concentration is regarded as the direct reflection of phytoplankton biomass, and widely used for measuring the degree of eutrophication of water bodies and marine ecosystems [1], [2], [3]. Chl-a data is also one of the typical ocean color products from remote sensing satellites such as the Moderate Resolution Imaging Spectroradiometer (MODIS) [4], Sea-Viewing Wide Field-of-View Sensor (SeaWiFS) [5], and Visible Infrared Imaging Radiometer Suite (VIIRS) [6] sensors. In general, the marine areas of high Chl-a are often prone to marine algal blooms such as red tides and green tides, which hinders the sustainable development of the marine environment and has significant impacts on the stability of marine ecosystems [7]. Chl-a thus plays an important role in monitoring and early warning of marine disasters, exploring the global carbon cycle, and analyzing the marine ecological environment.

However, the observed Chl-a data collected via satellites is often of high sparsity due to various unfavorable factors such as cloud cover, cloud adjacency effects, and thick aerosols [8], [9]. For example, the overall cloud-free coverage of MODIS-Terra and Aqua satellite over the ocean is only about 28% [10], and that of VIIRS satellite is about 30% [11]. Moreover, about 50% of the data in the cloud-free coverage areas is labeled as low quality data due to cloud adjacency effects [12]. Fig. 1 illustrates the spatiotemporal distributions of missing rates of Ocean Color Climate Change Initiative (OC-CCI) Chl-a data in the study area of the Yangtze River Estuary in 2022. OC-CCI Chl-a product fuses the satellite data from MODIS, VIIRS, SeaWiFS, and Medium Resolution Imaging Spectrometer (MERIS) sensors to produce daily Chl-a data. As shown in Fig. 1(a), the situation of daily data missing is severe, and the average missing rate is as high as 71.54%. As can be seen from Fig. 1(b), the missing rates for most areas are mainly concentrated between 70% and 80%, and the sparsity issue is particularly serious in the areas near the coastline. Therefore, the available Chl-a observation data is highly sparse in both temporal and spatial dimensions, which seriously influences its applications in practice, thus requiring effective data imputation methods to address this issue.

Chl-a data imputation can be formulated as a spatiotemporal imputation problem. The existing methods for Chl-a data imputation mainly falls into two categories, i.e., *statistical methods* and *machine learning methods*. Statistical methods utilize the statistical properties, e.g., linear correlations and simple nonlinear correlations, of the observed data to achieve data imputation. Representative statistical methods include optimal interpolation (OI) method [13], [14],

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Shuyu Wang, Wengen Li, Jihong Guan, Xiwei Liu, and Yichao Zhang are with the Department of Computer Science and Technology, Tongji University, Shanghai 200092, China (e-mail: tj_sywang@tongji.edu.cn; lwengen@tongji.edu.cn; jhguan@tongji.edu.cn; xwliu@tongji.edu.cn; yichaozhang@tongji.edu.cn).

Shuigeng Zhou is with the Department of Computer Science, Fudan University, Shanghai 200433, China (e-mail: sgzhou@fudan.edu.cn).

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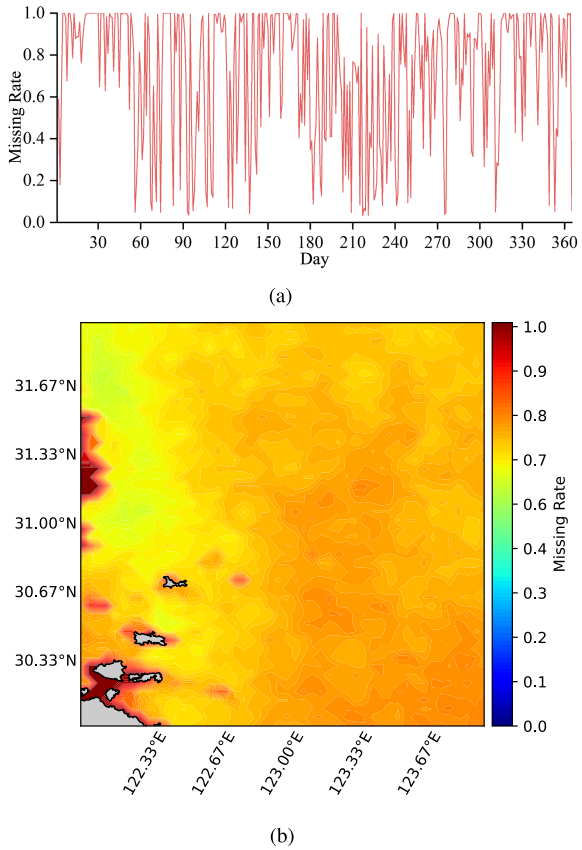


Fig. 1. Temporal and spatial distributions of the missing rates of OC-CCI Chl-a data in the region of Yangtze Estuary in 2022. (a) Daily missing rates of OC-CCI Chl-a data in 2022. (b) Spatial distribution of the missing rates of OC-CCI Chl-a data.

[15], [16], inverse distance weighted (IDW) method [17], Kriging [18], [19], and data interpolating empirical orthogonal functions (DINEOF) [20], [21], [22], [23]. However, it is usually difficult for statistical methods to capture the complex nonlinear spatiotemporal variations in Chl-a data to achieve accurate imputation.

With the rapid development of remote sensing technology, a vast amount of large-scale and long time-series Chl-a data have been accumulated. Chiu et al. [24] review the nature-inspired metaheuristic techniques for missing value imputation from 2011 to 2021, which indicates that researchers are more inclined to combine metaheuristic methods with machine learning methods to achieve missing value imputation. Therefore, Chl-a data imputation based on machine learning is becoming increasingly popular. Meanwhile, due to the high sparsity of Chl-a data, some studies introduce exogenous data, e.g., sea surface temperature (SST), photosynthetically active radiation (PAR), sea ice concentration (SIC), and sea surface height (SSH), to enrich the information for improving the accuracy of Chl-a data imputation. Chl-a imputation methods based on machine learning can be further categorized into *traditional machine learning methods* and *deep learning methods*. Traditional machine learning learns the hidden patterns in Chl-a data and exogenous data to achieve Chl-a data imputation, and the representative methods include random forest (RF), self-organizing map (SOM), artificial neural network (ANN), extremely randomized tree (ExtraTree), eXtreme gradient boosting (XGBoost), and support vector regression (SVR) [25], [26], [27], [28], [29], [30], [31]. For example, Park et al. [27] employed the ensemble learning methods RF and ExtraTree for Chl-a data imputation. They utilized cloud-free microwave and reanalysis datasets, including SST, SIC, PAR, and wind, to estimate the missing values of Chl-a, and found that SST and PAR in atmospheric components contribute significantly to the estimation. Then, they further introduced the Chl-a data from VIIRS to enhance Chl-a data imputation using RF [26]. Similarly, Xing et al. [28] used XGBoost to reconstruct the missing values in Chl-a data by introducing the reanalysis datasets of SST, PAR, SSH, and wind. Chen et al. [29] also used the RF method to fill the missing values of Chl-a data with the assistance of homogenous exogenous data, e.g., the ocean color index (CI). Jouini et al. [30] applied the SOM method to reconstruct sparse Chl-a data under strong cloud cover with the assistance of exogenous data, including SST and SSH. However, traditional machine learning methods can only learn knowledge from limited data, and cannot model the complex spatiotemporal dependencies and dynamic regularity in Chl-a data. Furthermore, these methods usually use reanalysis data as exogenous data, which may result in accumulated errors for Chl-a data imputation.

Deep learning methods are suited for time-series data imputation due to their excellent ability to learn the complex nonlinear correlations and underlying data distributions from large-scale data [32]. Barth et al. [33] proposed a Data-Interpolating Convolution Auto-Encoder (DINCAE 1.0) for AVHRR-SST, which is also used to reconstruct GlobColour project Chl-a data with cloud-free daily MW-IR SST in [34]. After that, DINCAE 2.0 was proposed for reconstructing MODIS-Terra SST and altimetry observations [35]. Long et al. [36] proposed a novel bishift network (BSN) model and introduced generative adversarial network (GAN) to reconstruct missing information with multitemporal remote sensing images. However, these methods do not account for the high sparsity of data, and ignore the spatial locations with a data missing rate exceeding a certain threshold, thus cannot work well on highly sparse data imputation.

Typical deep learning methods for continue time-series data imputation are mostly based on GAN and self-attention mechanisms. For example, GAIN [37], E^{2}GAN [38], and SolarGAN [39] are the typical GAN-based models for time-series data imputation, and they have excellent data generalization ability by learning the data distributions to generate data similar to the original data. Transformer [40] and its variants are self-attention-based methods that achieve high-quality data imputation by utilizing the nonautoregressive mechanism, and can effectively avoid compounding errors. These deep learning-based imputation methods for highly sparse continue time-series data can be directly applied to Chl-a data imputation and usually outperform statistical methods and traditional machine learning methods.

However, with the increase in missing rate, the data imputation performance of deep learning methods decreases drastically due to the limited available information about the missed records. Fig. 2 illustrates the Chl-a data imputation

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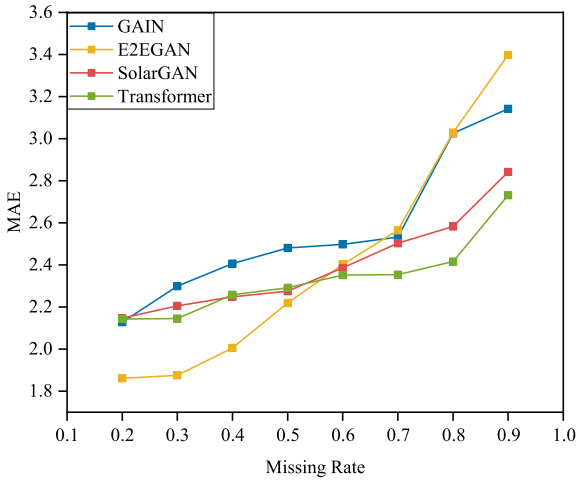


Fig. 2. Performance of deep learning-based data imputation methods for Chl-a data imputation.

performance of the four deep learning methods, i.e., GAIN, E^{2}GAN, SolarGAN, and Transformer, mentioned above in the Yangtze River Estuary area. When the missing rate reaches 0.7 or above, all these methods perform quite bad. The reason for this is that the high sparsity of Chl-a data implies that a large number of data samples have no ground truth, which results in very limited effective information for training the data imputation models. Naturally, due to insufficient training, the models fail to effectively learn the true distribution of Chl-a data, resulting in a significant bias between the generated data and the real data. In this work, to address the issue of high sparsity, we propose the S2-STI model for highly sparse Chl-a data imputation with multisource satellite data, including Chl-a data, SST data, and PAR data. In sum, the contributions of this work are summarized as follows.

1) We proposed the S2-STI model that could achieve effective imputation for highly sparse Chl-a data by fusing multisource satellite data.
2) We designed a multisource sparse data fusion (MSDF) module for fusing highly sparse Chl-a data and multisource lowly sparse exogenous data to learn high-quality spatiotemporal representations for Chl-a data imputation.
3) We developed a generation-based data imputation (GDI) module to learn the distribution of Chl-a data based on the outputs of MSDF module, and designed a self-supervised training strategy for S2-STI in the absence of ground truth.
4) We conducted extensive experiments on two real-word Chl-a datasets to evaluate the proposed method, and the results indicate that S2-STI outperforms the existing data imputation methods by a large margin.

The rest of this work is organized as follows. Section II describes the datasets and study areas. Section III details the proposed S2-STI model for highly sparse Chl-a data imputation. Then, we present the experiments and results in Section IV and conduct the discussion in Section V. Finally, we draw the conclusions in Section VI.

II. Data and Study Areas

A. Datasets

1) Chl-a Data: The Chl-a data is from OC-CCI 6.0 product[1] and collected once everyday. This product fuses the satellite data from MODIS, VIIRS, SeaWiFS, and MERIS sensors to produce daily Chl-a data. The spatial resolution is 4 km ($1/24° \times 1/24°$) and the recent time range is from September 1997 to December 2023.

We choose the SST from Advanced Very High Resolution Radiometer (AVHRR) satellite and PAR from MODIS-Terra satellite as the exogenous data since they both exhibit significant positive/negative correlations with Chl-a [41], [42], [43]. Fig. 3 presents the correlations between Chl-a and SST and PAR, respectively, and the correlations are measured using Pearson correlation coefficients. Fig. 3(a) demonstrates a significant negative correlation between Chl-a and SST with a correlation coefficient of $R = -0.69$. Meanwhile, Fig. 3(b) indicates a significant positive correlation between Chl-a and PAR with a correlation coefficient of $R = 0.33$. In general, both SST and PAR are also sparse but their missing rates are lower than that of Chl-a. Therefore, SST and PAR data could provide supplementary information for Chl-a data imputation, and their details are described as follows.

2) SST Data: The SST data is from AVHRR[2] and collected twice, i.e., daytime and nighttime, everyday. The spatial resolution is 4 km ($1/24° \times 1/24°$) and the recent time range is from August 1981 to September 2023.

3) PAR Data: The PAR data is from L3-level MODIS-Terra[3] and collected once everyday. The spatial resolution is 4km ($1/24° \times 1/24°$) and the time range is from February 2000 to present.

B. Study Areas

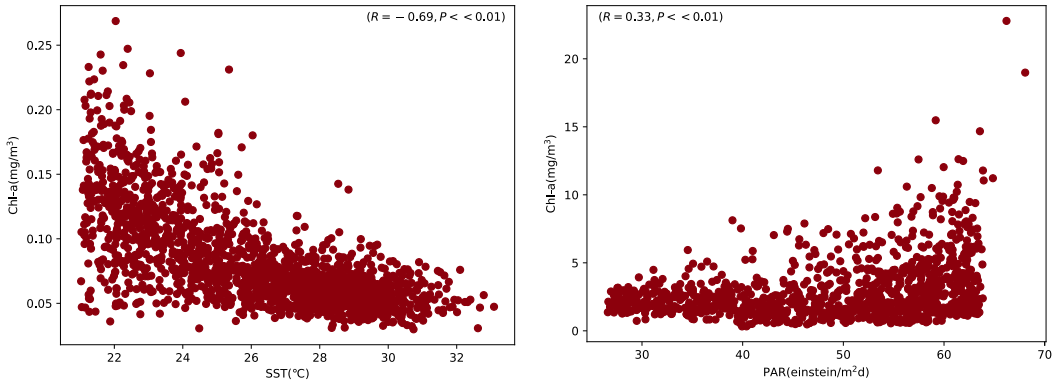
To unify the time range of the three datasets of Chl-a, SST, and PAR, we study the time range from February 25, 2000 to December 31, 2022. Two study areas, i.e., East China Sea and Yangtze River Estuary, near the land are selected since most ocean environment problems caused by Chl-a occur in coastal waters.

1) CHA-EAST: The study area of East China Sea covers the spatial region of [128.00°E–130.00°E, 24.00°N–26.00°N], and is divided into 48×48 grids of size $1/24° \times 1/24°$. The data missing rates of Chl-a, SST, and PAR in this area are 75.40%, 20.45%, and 21.03%, respectively. Fig. 4 illustrates the spatial distributions of Chl-a, SST, and PAR in CHA-EAST dataset on one selected day.

2) CHA-YRE: The study area of Yangtze River Estuary covers the spatial region of [122.00°E–124.00°E, 30.00°N–32.00°N], and is also divided into 48×48 grids of size $1/24° \times 1/24°$. The data missing rates of Chl-a, SST, and PAR in this area are 83.85%, 33.61%, and 6.65%, respectively. Fig. 5 illustrates the spatial distributions of Chl-a, SST, and PAR in CHA-YRE on a selected day. Compared with CHA-EAST dataset, the Chl-a in CHA-YRE dataset exhibits

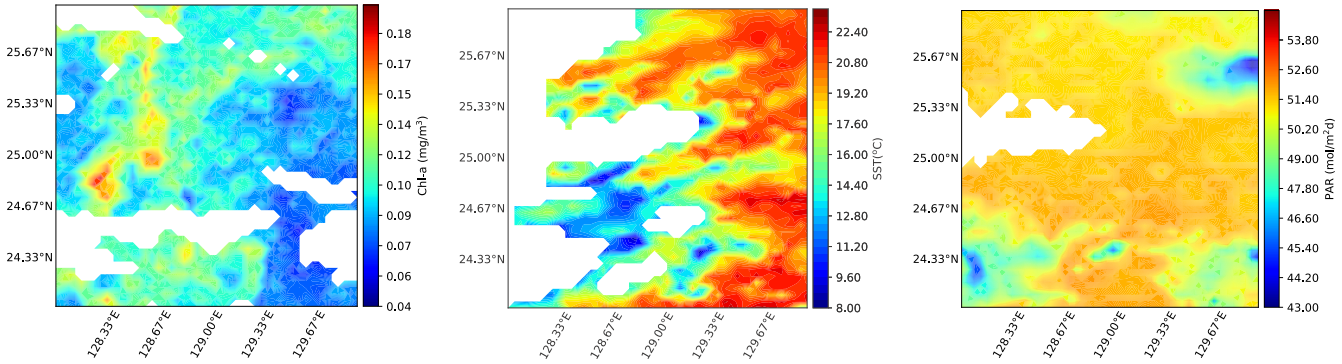
[1] https://esa-oceancolour-cci.org/
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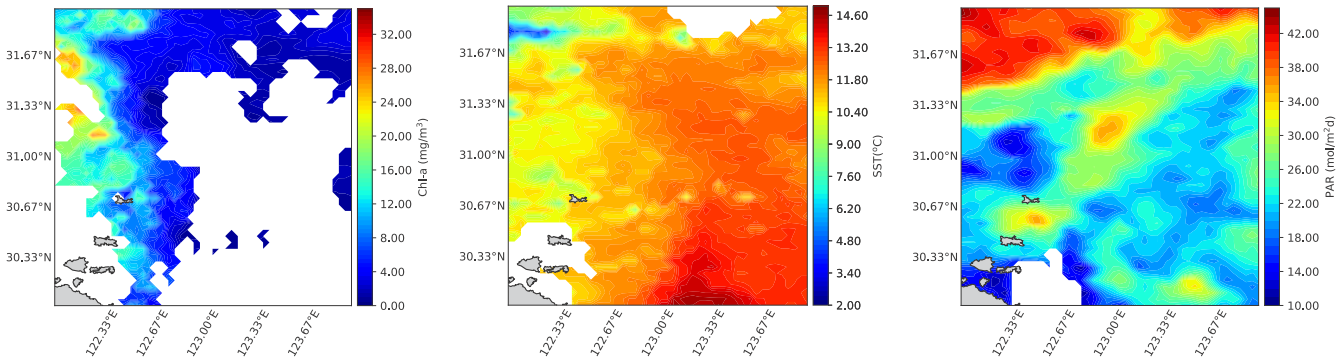
(a) (b)

Fig. 3. Correlations between Chl-a and the two selected exogenous data sources, i.e., SST and PAR. (a) Chl-a versus SST. (b) Chl-a versus PAR.



(a) (b) (c)

Fig. 4. Spatial distributions of Chl-a, SST, and PAR in CHA-EAST dataset on March 16, 2022, where the white areas have no data. (a) OC-CCI Chl-a. (b) AVHRR SST. (c) MODIS-Terra PAR.



(a) (b) (c)

Fig. 5. Spatial distributions of Chl-a, SST, and PAR in CHA-YRE dataset on March 16, 2022, where the white areas have no data, and the gray areas are land. (a) OC-CCI Chl-a. (b) AVHRR SST. (c) MODIS-Terra PAR.

significant fluctuations since Yangtze River Estuary is more close to the coastline.

III. Method

A. Problem Definition

Chl-a data imputation is usually formulated as a spatiotemporal imputation problem, i.e., learning the spatiotemporal distribution of Chl-a data based on the observed Chl-a data and exogenous auxiliary data to reconstruct the missing data. In this work, we divide the data used for Chl-a data imputation into two categories, i.e., target data and exogenous data, where target data corresponds to the highly sparse Chl-a data that requires to be imputed, and exogenous data corresponds to the SST data and PAR data that have a lower missing rate than the target data and are used to assist the imputation of target data.

As shown in Fig. 6, remote sensing satellite data is typically represented as grid data. Specifically, Chl-a data is denoted as $\mathcal{Y} = \{y_1, y_2, \ldots, y_T\} \in \mathbb{R}^{T \times N}$, where T is the number of time steps, N is the number of grid regions, and y_t $(1 \leq t \leq T)$ records the Chl-a observations for N locations at time step t.

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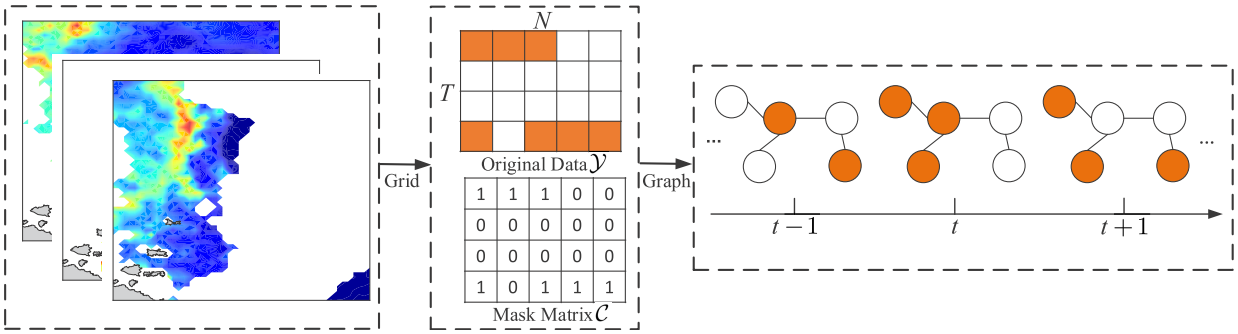


Fig. 6. Chl-a remote sensing data sequence, and its representations using matrix and dynamic graph, respectively, where orange color regions represent the nodes with observed values, and white color regions represent the nodes with missing values.

Chl-a data matrix $\mathcal{Y}$ is highly sparse and the corresponding mask matrix is $\mathcal{C}$. The values in $\mathcal{C}$ are determined as follows:

$$c_{t,i} = \begin{cases} 1, & \text{if } y_{t,i} \text{ is observed} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

To capture the spatial correlations between grid regions, we construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, where the node set $\mathcal{V}$ corresponds to the N grid regions, i.e., $|\mathcal{V}| = N$, $\mathcal{E}$ is the set of edges, and two nodes are connected if their distance is within a certain threshold. The adjacency matrix $\mathcal{A} \in \mathbb{R}^{N\times N}$ records the connections between nodes, and each item $a_{i,j}$ in $\mathcal{A}$ is determined as follows:

$$a_{i,j} = \begin{cases} 1, & \text{if } \exp\left\{\dfrac{-d_{i,j}}{\eta}\right\}^2 \geq \psi \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $d_{i,j}$ is the spatial distance between nodes v_i and v_j, η is a scaling factor, and ψ is the threshold that controls the sparsity of matrix $\mathcal{A}$.

Let $\mathcal{X} = \{X^1, X^2, \ldots, X^F\} \in \mathbb{R}^{T\times N\times F}$ denote the exogenous data of F types, where $X^f = \{x_1^f, x_2^f, \ldots, x_T^f\} \in \mathbb{R}^{T\times N}$ $(f = 1, 2, \ldots, F)$ is a time series for the fth type of exogenous data of N locations with T time steps. We also introduce a mask matrix $M^f \in \mathbb{R}^{T\times N}$ for X^f, and the way to determine the value of item $m_{t,i}^f$ in M^f is similar to $c_{t,i}$ in (1).

For the Chl-a data imputation problem, based on the target Chl-a time series $\mathcal{Y}$, the corresponding mask matrix $\mathcal{C}$, the data graph structure $\mathcal{G}$, and the exogenous data $\mathcal{X}$, we aim to find a function $\mathcal{F}$ to fill the missing values in $\mathcal{Y}$ to obtain the completed sequence $\widetilde{\mathcal{Y}}$, i.e.,

$$\widetilde{\mathcal{Y}} = \mathcal{F}(\mathcal{Y}, \mathcal{C}, \mathcal{G}; \mathcal{X}). \tag{3}$$

B. Model Overview

Fig. 7 shows the architecture of S2-STI model which consists of the MSDF module and GDI module. First, the MSDF module fuses the matrix $\overline{\mathcal{Y}}$ of the randomly masked Chl-a data and the spatiotemporal representation matrix E of SST and PAR obtained by the spatiotemporal graph network (STGN). Then, the GDI module learns the underlying spatiotemporal distribution of Chl-a data based on the outputs of the fusion module via a GAN. Considering the high sparsity of Chl-a data, we designed a self-supervised training strategy to train S2-STI in the absence of ground truth. Finally, the missing values in the highly sparse Chl-a data $\mathcal{Y}$ are filled according to the data generated by the trained S2-STI model, i.e., $\widehat{\mathcal{Y}}$.

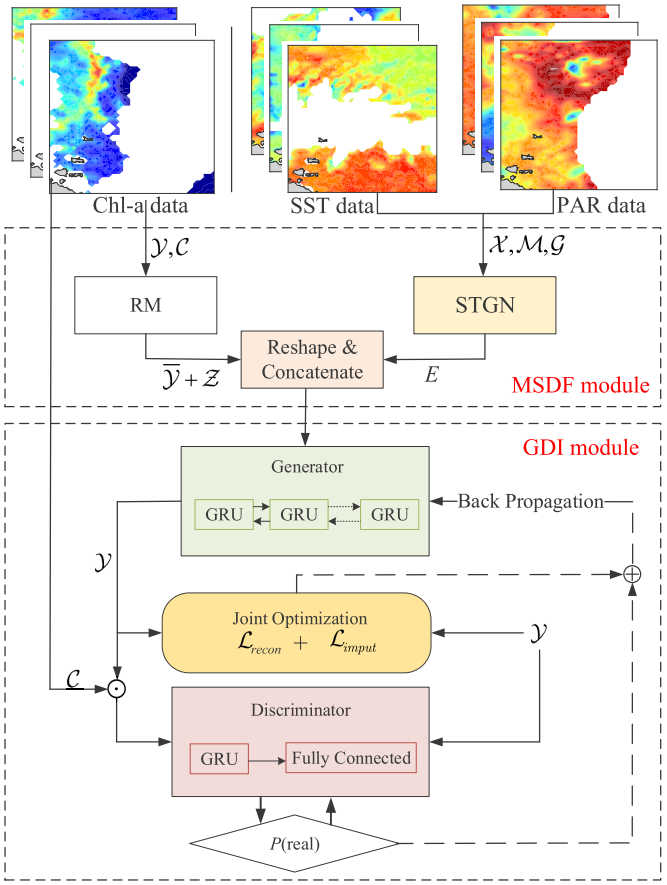


Fig. 7. Architecture of S2-STI model which consists of the MSDF module and GDI module, where RM is randomly mask operation for highly sparse Chl-a data $\mathcal{Y}$, and $\odot$ represents Hadamard product.

C. Multisource Sparse Data Fusion

The MSDF module first randomly masks a certain percentage of the original observed Chl-a values $\mathcal{Y}$ as the imputation target and obtains the residual data $\overline{\mathcal{Y}}$. We introduce a random noise variable $\mathcal{Z}$ for the nodes with missing values of the residual data $\overline{\mathcal{Y}}$ and thus obtain $\overline{\mathcal{Y}}+\mathcal{Z}$. Then, MSDF learns the

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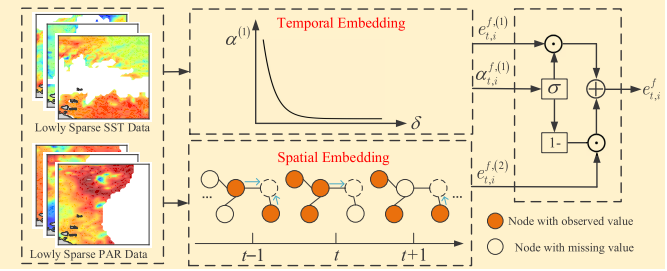


Fig. 8. Structure of STGN module.

spatiotemporal representations $\boldsymbol{E}$ of SST and PAR using the STGN network. After that, MSDF reshapes $\overline{\boldsymbol{\mathcal{Y}}} + \boldsymbol{\mathcal{Z}} \in \mathbb{R}^{T\times N}$ into the space of $\mathbb{R}^{T\times N\times 1}$, and concatenates it with matrix $\boldsymbol{E} \in \mathbb{R}^{T\times N\times F}$ to obtain the comprehensive representations about Chl-a. The fused results are then used for Chl-a data imputation in the GDI module.

Concretely, let $\overline{\boldsymbol{\mathcal{C}}}$ denote the randomly mask matrix corresponding to $\boldsymbol{\mathcal{C}}$, and each item $\overline{c}_{t,i}$ in $\overline{\boldsymbol{\mathcal{C}}}$ is determined as follows:

$$\overline{c}_{t,i} = \begin{cases} 0, & \text{if } c_{t,i} = 0 \mid y_{t,i} \text{ is randomly masked} \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

For exogenous data, e.g., SST and PAR, we first learn their embedding representations separately based on the spatial correlation graph $\boldsymbol{\mathcal{G}}$. Since SST and PAR data are also sparse, those nodes, i.e., grid regions, are embedded in different ways. For those nodes with observed values, we use the nonlinear activation function rectified linear unit (ReLU) to embed them, i.e.,

$$e_{t,i}^{f} = \text{ReLU}\left(x_{t,i}^{f}\right) \tag{5}$$

where $x_{t,i}^{f}$ represents the value of the fth type exogenous data of node v_i at time step t.

For those nodes without observed values, the observed values of neighboring nodes are used to generate their embedding representations according to the spatiotemporal correlations. As illustrated in Fig. 8, temporal embedding and spatial embedding are introduced to learn the spatiotemporal correlations between nodes in graph $\boldsymbol{\mathcal{G}}$.

1) Temporal Embedding: For the target node v_i with missing value at time step t, temporal embedding is achieved based on the nearest observation value $L_{t,i}^{f}$ in history. We define $\delta_{t,i}^{f}$ as the time interval from the nearest history time step with observation value to time step t, i.e.,

$$\delta_{t,i}^{f} = \begin{cases} 1 + \delta_{t-1,i}^{f}, & t > 0 \ \ \& \ \ m_{t,i}^{f} = 0 \\ 1, & t > 0 \ \ \& \ \ m_{t,i}^{f} = 1 \\ 0, & t = 0. \end{cases} \tag{6}$$

Since the missing values in exogenous data $\boldsymbol{\mathcal{X}}$ are random, we further define the time decay factor $\alpha_{t,i}^{f,(1)}$ based on $\delta_{t,i}^{f}$ as the weight of $L_{t,i}^{f}$, i.e.,

$$\alpha_{t,i}^{f,(1)} = \exp\left\{-\max\left(0, w_{t,i}^{f}(\alpha)\delta_{t,i}^{f} + b_{t,i}^{f}(\alpha)\right)\right\} \tag{7}$$

where $w_{t,i}^{f}(\alpha)$ and $b_{t,i}^{f}(\alpha)$ are learnable parameters.

In addition, considering that target node v_i may have no observation values of the fth type exogenous data for a long time, we take the mean value $e_{i,\text{mean}}^{f,(1)}$ of all the observed values of v_i into account. Then, the temporal embedding for the fth type exogenous data of node v_i at time step t is defined as follows:

$$e_{t,i}^{f,(1)} = \alpha_{t,i}^{f,(1)} L_{t,i}^{f} + \left(1 - \alpha_{t,i}^{f,(1)}\right) e_{i,\text{mean}}^{f,(1)} \tag{8}$$

where $\alpha_{t,i}^{f,(1)}$ is calculated via (7).

2) Spatial Embedding: For the target node v_i with missing value at time step t, spatial embedding is achieved based on the neighbor nodes with observed values. Since the missing status of data is random at each time step, we introduce the concept of dynamic adjacency matrix to represent the neighbor nodes with observed values at each time step. Concretely, based on the adjacency matrix $\boldsymbol{\mathcal{A}}$ and mask matrix $\boldsymbol{M}^{f}$, the item in the dynamic adjacency matrix $\widehat{\boldsymbol{A}}^{f} \in \mathbb{R}^{T\times N\times N}$ is defined as follows:

$$\widehat{a}_{t,i,j}^{f} = \begin{cases} 1, & \text{if } m_{t,j}^{f} = 1 \ \ \& \ \ a_{i,j} = 1 \\ 0, & \text{otherwise}. \end{cases} \tag{9}$$

Therefore, node v_j is a dynamic neighbor node of node v_i only if it is a neighbor node of v_i (i.e., $a_{i,j} = 1$) and has the observed value at time step t (i.e., $m_{t,j}^{f} = 1$).

To generate the spatial embedding of node v_i, we define the weight corresponding to the fth type exogenous data of node v_i at time step t as $\alpha_{t,i}^{f,(2)}$, i.e.,

$$\alpha_{t,i}^{f,(2)} = |\widehat{A}_{t,i}^{f}|/|A_{i}^{f}| \tag{10}$$

where $|\widehat{A}_{t,i}^{f}|$ and $|A_{i}^{f}|$ are the numbers of dynamic neighbor nodes and neighbor nodes, respectively, for node v_i at time step t when considering the fth type of exogenous data.

Then, the spatial embedding of node v_i is defined as follows:

$$e_{t,i}^{f,(2)} = \alpha_{t,i}^{f,(2)} s_{t,i}^{f} + \left(1 - \alpha_{t,i}^{f,(2)}\right) e_{i,\text{mean}}^{f,(2)} \tag{11}$$

where $e_{i,\text{mean}}^{f,(2)}$ is the mean value of the observed values of all the neighbor nodes except node v_i, and $s_{t,i}^{f}$ is calculated based on the set of dynamic neighbor nodes $\widehat{N}_{t,i}^{f}$ for node v_i of the fth type exogenous data at time step t, i.e.,

$$s_{t,i}^{f} = \frac{1}{|\widehat{A}_{t,i}^{f}|} \sum_{j\in\widehat{N}_{t,i}^{f}} e_{t,j}^{f}. \tag{12}$$

Finally, we obtain the embedding representation $e_{t,i}^{f}$ for node v_i of the fth type of exogenous data at time step t by performing a weighted sum of the temporal embedding $e_{t,i}^{f,(1)}$ and spatial embedding $e_{t,i}^{f,(2)}$, i.e.,

$$e_{t,i}^{f} = \beta_{t,i}^{f} e_{t,i}^{f,(1)} + \left(1 - \beta_{t,i}^{f}\right) e_{t,i}^{f,(2)} \tag{13}$$

where $\beta_{t,i}^{f} \in [0, 1]$ represents the dynamic weight for balancing temporal embedding $e_{t,i}^{f,(1)}$ and spatial embedding $e_{t,i}^{f,(2)}$, and is defined as follows:

$$\beta_{t,i}^{f} = \sigma\left(w_{t,i}^{f}(\beta)\alpha_{t,i}^{f,(1)} + b_{t,i}^{f}(\beta)\right) \tag{14}$$

where $w_{t,i}^{f}(\beta)$ and $b_{t,i}^{f}(\beta)$ are learnable parameters.

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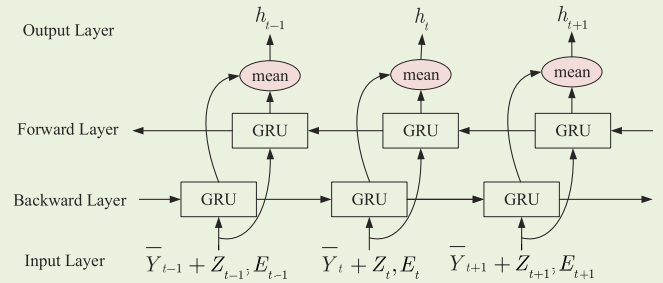


Fig. 9. Structure of the generator G in GDI module.

In sum, for the fth type of exogenous data, we first obtain the spatiotemporal representation matrix $\boldsymbol{E}^f$ for N locations at all T time steps, i.e., $\boldsymbol{E}^f = \{e_1^f, e_2^f, \ldots, e_T^f\} \in \mathbb{R}^{T\times N}$ $(f = 1, 2, \ldots, F)$ using the STGN module. Then, we combine $\boldsymbol{E}^1, \ldots, \boldsymbol{E}^F$ to obtain the spatiotemporal representation matrix $\boldsymbol{E} \in \mathbb{R}^{T\times N\times F}$ for all exogenous data.

D. Generation-Based Data Imputation

GDI module introduces Wasserstein GAN (WGAN) [44] to reconstruct the missing values in sparse Chl-a satellite data, and redesigns the structure of the generator and discriminator to better learn the distribution of Chl-a data by incorporating exogenous data. Gated recurrent unit (GRU) [45], a state-of-the-art RNN unit, has powerful capability in modeling temporal dependency. Therefore, we choose GRU as the network structure for the generator and discriminator. Fig. 9 shows the structure of the generator G of GDI module that consists of a sequence of bidirectional GRU units.

The updating functions of GRU in generator G are defined as follows:

$$\boldsymbol{r}_t = \sigma\left(\boldsymbol{W}_r\left[\overline{\boldsymbol{Y}}_t + \boldsymbol{Z}_t, \boldsymbol{E}_t, \boldsymbol{h}_t\right] + \boldsymbol{b}_r\right) \tag{15}$$

$$\boldsymbol{u}_t = \sigma\left(\boldsymbol{W}_u\left[\overline{\boldsymbol{Y}}_t + \boldsymbol{Z}_t, \boldsymbol{E}_t, \boldsymbol{h}_t\right] + \boldsymbol{b}_u\right) \tag{16}$$

$$\widehat{\boldsymbol{h}}_t = \tanh\left(\boldsymbol{W}_{\widehat{h}}\left[\overline{\boldsymbol{Y}}_t + \boldsymbol{Z}_t, \boldsymbol{E}_t, \boldsymbol{h}_{t-1} \odot \boldsymbol{r}_t\right] + \boldsymbol{b}_{\widehat{h}}\right) \tag{17}$$

$$\boldsymbol{h}_t = (1 - \boldsymbol{u}_t) \odot \boldsymbol{h}_{t-1} + \boldsymbol{u}_t \odot \widehat{\boldsymbol{h}}_j \tag{18}$$

where $\boldsymbol{r}_t$ is the reset gate used to determine the forgotten information of each GRU unit, $\boldsymbol{u}_t$ is the update gate to control the output information, $\widehat{\boldsymbol{h}}_t$ is the candidate hidden state, $\boldsymbol{h}_t$ is the output of GRU, $\sigma(\cdot)$ is the sigmod function, $\tanh(\cdot)$ is the hyperbolic tangent function, $\odot$ is the Hadamard product, $\boldsymbol{W}_r$, and $\boldsymbol{W}_u$, $\boldsymbol{W}_{\widehat{h}}$, $\boldsymbol{b}_r$, $\boldsymbol{b}_u$, and $\boldsymbol{b}_{\widehat{h}}$ are learnable parameters.

For the bidirectional GRU unit, we perform a mean operation on the hidden states obtained from the forward layer and the backward layer to obtain the final hidden state $\boldsymbol{h}_t$ at time step t.

The discriminator D consists of unidirectional GRU units, and takes the original data matrix $\boldsymbol{\mathcal{Y}}$ and the matrix $\boldsymbol{\mathcal{C}} \odot \widehat{\boldsymbol{\mathcal{Y}}}$ as inputs to distinguish that the $\widehat{\boldsymbol{\mathcal{Y}}}$ at nodes with observed values are fake or real.

The loss function $\mathcal{L}_G$ of generator G is designed as follows:

$$\mathcal{L}_G = \lambda\left(\mathcal{L}_{\text{imput}} + \mathcal{L}_{\text{recon}}\right) - D\left(\boldsymbol{\mathcal{C}} \odot \widehat{\boldsymbol{\mathcal{Y}}}\right) \tag{19}$$

where $D(\boldsymbol{\mathcal{C}} \odot \widehat{\boldsymbol{\mathcal{Y}}})$ measures the probability that indicates the degree of authenticity of the generated Chl-a data $\widehat{\boldsymbol{\mathcal{Y}}}$ at nodes with observed values, $\mathcal{L}_{\text{imput}}$ is the imputation loss, $\mathcal{L}_{\text{recon}}$ is the reconstruction loss, and λ is the hyperparameter that balances $\mathcal{L}_{\text{imput}} + \mathcal{L}_{\text{recon}}$ and $D(\boldsymbol{\mathcal{C}} \odot \widehat{\boldsymbol{\mathcal{Y}}})$.

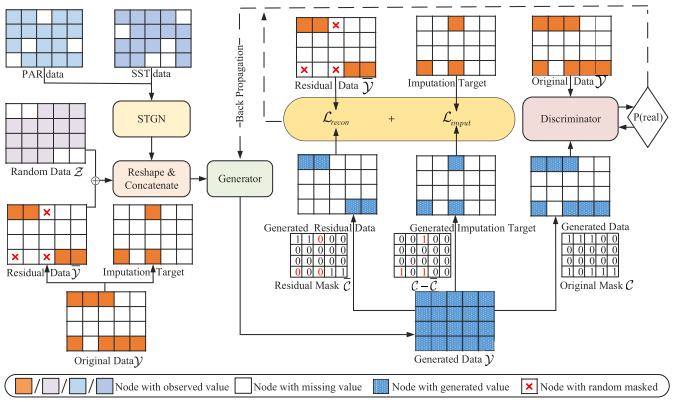


Fig. 10. Self-supervised training strategy for S2-STI model.

The imputation loss $\mathcal{L}_{\text{imput}}$ and reconstruction loss $\mathcal{L}_{\text{recon}}$ guide the generator G to generate data that follows the original distribution of Chl-a data, where $\mathcal{L}_{\text{imput}}$ measures the difference between the masked data and the generated data, and $\mathcal{L}_{\text{recon}}$ measures the difference between the residual observed values after masking and the corresponding generated data, i.e.,

$$\mathcal{L}_{\text{imput}} = \left\|\left(\boldsymbol{\mathcal{C}} - \overline{\boldsymbol{\mathcal{C}}}\right) \odot \boldsymbol{\mathcal{Y}} - \left(\boldsymbol{\mathcal{C}} - \overline{\boldsymbol{\mathcal{C}}}\right) \odot \widehat{\boldsymbol{\mathcal{Y}}}\right\|_2 \tag{20}$$

$$\mathcal{L}_{\text{recon}} = \left\|\overline{\boldsymbol{\mathcal{C}}} \odot \boldsymbol{\mathcal{Y}} - \overline{\boldsymbol{\mathcal{C}}} \odot \widehat{\boldsymbol{\mathcal{Y}}}\right\|_2. \tag{21}$$

The discriminator D is designed to distinguish the real Chl-a data $\boldsymbol{\mathcal{Y}}$ and the generated matrix $\widehat{\boldsymbol{\mathcal{Y}}}$, and the corresponding loss function is

$$\mathcal{L}_D = D\left(\boldsymbol{\mathcal{C}} \odot \widehat{\boldsymbol{\mathcal{Y}}}\right) - D(\boldsymbol{\mathcal{Y}}) \tag{22}$$

where $D(\mathrm{C} \odot \widehat{Y})$ is the same as that in the loss function of generator G, and $D(\mathrm{Y})$ measures the probability that indicates the degree of authenticity of the ground truth of Chl-a data.

E. Self-Supervised Training Strategy for S2-STI Model

Inspired by the idea of the masked autoencoder (MAE) [46], we designed a self-supervised training strategy, as shown in Fig. 10, to address the issue of lacking of ground truth in the highly sparse Chl-a dataset. Specifically, we validate the imputation performance of S2-STI model by masking a portion of the original observed values and then reconstructing the masked values.

Algorithm 1 presents the pseudocode for training S2-STI model. Finally, we fill the missing values in the sparse Chl-a data matrix $\boldsymbol{\mathcal{Y}}$ with the generated data matrix $\widehat{\boldsymbol{\mathcal{Y}}}$ to obtain the completed sequence $\widetilde{\boldsymbol{\mathcal{Y}}}$, i.e.,

$$\widetilde{\boldsymbol{\mathcal{Y}}} = \boldsymbol{\mathcal{Y}} \odot \boldsymbol{\mathcal{C}} + (\mathbf{1} - \boldsymbol{\mathcal{C}}) \odot \widehat{\boldsymbol{\mathcal{Y}}}. \tag{23}$$

IV. Experiments and Results

A. Experimental Settings

1) Baseline Methods: To evaluate the effectiveness of S2-STI model, we compare it with multiple baseline methods, including the following.

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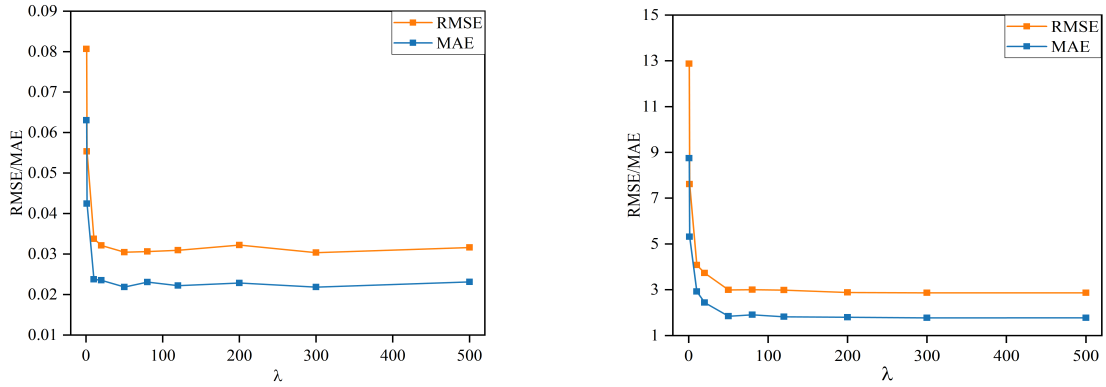


Fig. 11. Results of S2-STI model while varying the parameter λ, where the missing rate of Chl-a data is 0.5. (a) CHA-EAST. (b) CHA-YRE.

Algorithm 1 Training Strategy of S2-STIModel

Input: highly sparse Chl-a data $\mathcal{Y}$, lowly sparse exogenous data $\mathcal{X}$, the number of epochs N_{epoch}, noise matrix $\mathcal{Z}$, and mask matrices $\mathcal{C}$ and $\mathcal{M}$

Output: Generated Chl-a data matrix $\widehat{\mathcal{Y}}$

$epoch \leftarrow 0$
while $epoch \leq N_{epoch}$ **do**
  Obtain the spatiotemporal graph embedding $\boldsymbol{E}$ by STGN module
  Randomly mask a portion of $\mathcal{Y}$ to obtain the imputation targets and residual data $\overline{\mathcal{Y}}$
  Obtain the generated matrix $\widehat{\mathcal{Y}} \leftarrow G(\overline{\mathcal{Y}} + \mathcal{Z}, \boldsymbol{E})$
  Calculate the loss functions $\mathcal{L}_{\text{imput}}$ and $\mathcal{L}_{\text{recon}}$, i.e., $\mathcal{L}_{\text{imput}} \leftarrow \|(\mathcal{C} - \overline{\mathcal{C}}) \odot \mathcal{Y} - (\mathcal{C} - \overline{\mathcal{C}}) \odot \widehat{\mathcal{Y}}\|_2$, $\mathcal{L}_{\text{recon}} \leftarrow \|\overline{\mathcal{C}} \odot \mathcal{Y} - \overline{\mathcal{C}} \odot \widehat{\mathcal{Y}}\|_2$
  Calculate the loss of G, i.e., $\mathcal{L}_G \leftarrow \lambda(\mathcal{L}_{\text{imput}} + \mathcal{L}_{\text{recon}}) - D(\mathcal{C} \odot \widehat{\mathcal{Y}})$
  Back-propagate $\mathcal{L}_G$ to update G
  Calculate the loss of D, i.e., $\mathcal{L}_D \leftarrow D(\mathcal{C} \odot \widehat{\mathcal{Y}}) - D(\mathcal{Y})$
  Back-propagate $\mathcal{L}_D$ to update D
end while
return Generated Chl-a data matrix $\widehat{\mathcal{Y}}$

1) *MEAN:* It achieves data imputation by using the mean values of historical records.
2) *LAST:* The latest observed value in the historical records is used as the imputed value.
3) *GAIN [37]:* A GAN-based data imputation method that introduces a hinting mechanism in discriminator to distinguish the original data and the generated data.
4) *E²GAN [38]:* An end-to-end imputation method based on GAN, where the generator achieves data imputation by compressing and reconstructing the autoencoder structure.
5) *SolarGAN [39]:* A GAN-based imputation method for relatively independent time series by fusing the noise matrix and data matrix as the inputs of the generator.
6) *Transformer [40]:* An attention-based method that simultaneously focusses on the information at different positions in the sequence through a multihead self-attention mechanism to comprehensively capture the characteristics of time series.
7) *TimesNet [47]:* A multiperiodicity perspective time-series analysis method that captures complex temporal features by transforming 1-D time series into a 2-D space tensor.
8) *iTransformer [48]:* An attention-based method that captures location correlations by inverting the transformer to treat each single location time sequence as a token.
9) *DINCAE [33]:* A convolutional neural network-based method that utilizes the autoencoder structure to learn the latent representation of Chl-a data and achieve data imputation.

2) Evaluation Metrics: Three evaluation metrics, i.e., root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), are utilized to evaluate the performance of imputation methods, i.e.,

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^{T}\left(\left(y_t - \widehat{y}_t\right) \odot \left(c_t - \overline{c}_t\right)\right)^2}{\sum_{t=1}^{T} c_t - \overline{c}_t}} \tag{24}$$

$$\text{MAE} = \frac{\sum_{t=1}^{T}\left|\left(y_t - \widehat{y}_t\right) \odot \left(c_t - \overline{c}_t\right)\right|}{\sum_{t=1}^{T} c_t - \overline{c}_t} \tag{25}$$

$$\text{MAPE} = \frac{\sum_{t=1}^{T}\left|\left(y_t - \widehat{y}_t\right) \odot \left(c_t - \overline{c}_t\right)\right|}{\sum_{t=1}^{T}\left|y_t \odot \left(c_t - \overline{c}_t\right)\right|} \tag{26}$$

where y_t and $\widehat{y}_t$ are the true value and generated value at time step t for all nodes, and c_t and $\overline{c}_t$ are the corresponding mask for the true value y_t and the residual value $\overline{y}_t$ at time step t, respectively. The missing values in the datasets are not evaluated due to the absence of ground truth.

3) Training and Testing Datasets: For all experiments, each dataset is divided chronologically into training, validation and testing datasets. Concretely, we use 80% of each dataset for training, and the remaining 20% is equally divided into validation, and testing datasets.

B. Hyperparameter Studies

We conduct experiments to determine the optimal values of the hyperparameters, including the balance factor λ, the

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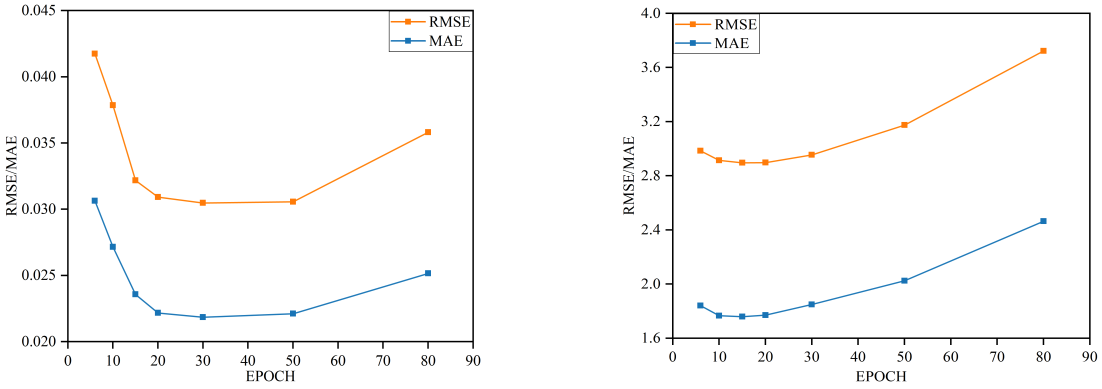


Fig. 12. Results of S2-STI model while varying the number of training epochs, where the missing rate of Chl-a data is 0.5. (a) CHA-EAST. (b) CHA-YRE.

number of training epochs N_{epoch}, and the threshold ψ, for S2-STI model.

1) Influence of the Parameter λ*:* The hyperparameter λ is used to balance the discriminator's output probability and the joint loss (i.e., imputation loss and reconstruction loss), and is crucial for the imputation results. Fig. 11 shows the RMSE and MAE of S2-STI while varying λ, where the Chl-a data missing rate is set to 0.5. It is worth mentioning that the missing rate here is the proportion of randomly masked observations on the original sparse dataset. According to the results, with the increase in λ, the performance of S2-STI first increases sharply and then keeps stable. Accordingly, we set $\lambda = 50$ for CHA-EAST dataset and $\lambda = 200$ for CHA-YRE dataset.

2) Influence of the Number of Training Epochs: The appropriate number of epochs is crucial for training S2-STI model. Too few epochs usually lead to underfitting, and the data imputation model will fail to capture the complex correlations in the original data. Too many epochs will lead to overfitting, resulting in poor generalization of the model. Fig. 12 demonstrates the MAE and RMSE of S2-STI while varying the number of training epochs. According to the results, we set the number of epochs to 30 for CHA-EAST dataset and 15 for CHA-YRE dataset.

3) Influence of the Threshold ψ*:* The adjacency matrix $\boldsymbol{\mathcal{A}}$ records the neighboring nodes for each node. Too few neighboring nodes cannot fully capture the spatial dependencies in Chl-a data. In contrast, too many neighboring nodes would introduce excessive noise, leading to information bias. In S2-STI model, the number of neighboring nodes is determined by parameters η and ψ according to (2). Based on the spatial resolutions and study areas of the datasets, we first set η to 100 to ensure that the number of neighboring nodes does not exceed half the total number of nodes. After fixing η, we evaluate the performance of S2-STI while varying ψ to determine the optimal settings of ψ. Fig. 13 shows the RMSE and MAE of S2-STI while varying ψ, where the missing rate of Chl-a data is set to 0.5. The results show that $\psi = 0.4$ and $\psi = 0.6$ are suitable values for CHA-EAST and CHA-YRE datasets, respectively.

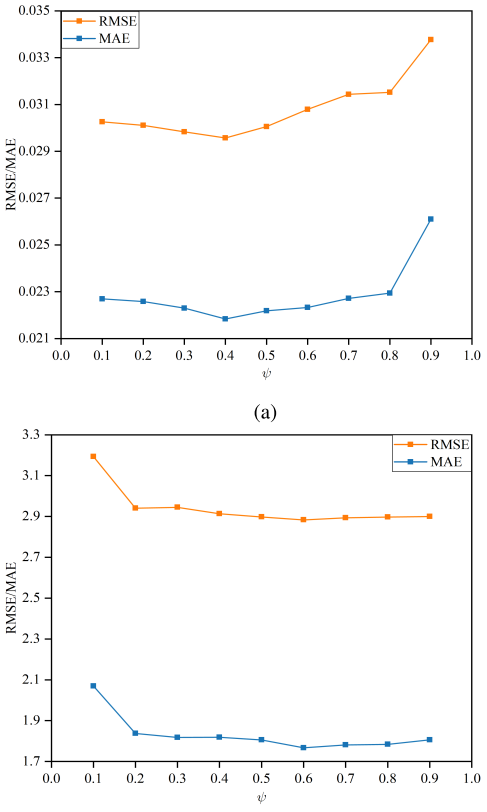


Fig. 13. Results of S2-STI model while varying ψ, where the missing rate of Chl-a data is 0.5. (a) CHA-EAST. (b) CHA-YRE.

C. Experimental Results and Analysis

Tables I and II show the data imputation results, including RMSE, MAE, and MAPE, on the CHA-EAST and CHA-YRE datasets, respectively, while varying the missing rate. According to the results, S2-STI largely outperforms the existing imputation methods, especially for high missing rates. Specifically, when the missing rate reaches 0.8 or above, S2-STI outperforms the second best results by more than 1.48% in terms of RMSE, and by 10% in terms of MAE and MAPE on the CHA-EAST dataset. As for the CHA-YRE dataset, the

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TABLE I

RESULTS OF S2-STIAND BASELINE DATA IMPUTATION METHODS ON THE CHA-EAST DATASET WHILE VARYING THE MISSING RATE, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND TEXTCOLOR[RGB]0.55, 0, 0↑ REPRESENTS THE MINIMUM INCREASE RATIO, ↓ REPRESENTS THE MAXIMUM DECREASE RATIO

Method	0.2			0.3			0.4			0.5		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
MEAN	0.0829	0.0689	75.55%	0.0829	0.0689	75.54%	0.0829	0.0689	75.54%	0.0829	0.0689	75.55%
LAST	0.0702	0.0462	46.20%	0.0728	0.0491	49.10%	0.0756	0.0523	53.95%	0.0788	0.0562	56.30%
GAIN	0.0345	0.0253	27.64%	0.0364	0.0264	28.88%	0.0369	0.0279	30.50%	0.0386	0.0286	31.41%
E^2GAN	0.0467	0.0328	35.77%	0.0487	0.0343	37.13%	0.0502	0.0357	39.98%	0.0542	0.0371	40.52%
SolarGAN	0.0298	0.0203	22.94%	0.0300	0.0213	23.43%	0.0315	0.0225	24.38%	0.0322	0.0226	24.96%
Transformer	0.0323	0.0248	27.38%	0.0333	0.0257	28.42%	0.0351	0.0273	30.15%	0.0364	0.0283	31.31%
TimesNet	0.0279	0.0206	22.71%	0.0285	0.0207	**22.84%**	0.0308	0.0238	26.30%	0.0316	0.0244	26.94%
iTransformer	0.0290	0.0195	22.59%	0.0297	**0.0198**	22.99%	0.0310	0.0212	24.56%	0.0315	0.0219	25.24%
DINCAE	**0.0244**	**0.0183**	**20.52%**	0.0288	0.0208	23.07%	0.0297	**0.0211**	23.39%	0.0306	0.0238	26.30%
S2-STI	0.0275	0.0203	22.84%	**0.0284**	0.0209	23.00%	**0.0289**	**0.0211**	**23.19%**	**0.0296**	**0.0218**	**24.09%**
Δ	12.70%↓	10.93%↓	11.31%↓	0.35%↑	5.56%↓	0.70%↓	2.69%↑	0%↑	0.86%↑	3.27%↑	0.46%↑	3.49%↑
Method	0.6			0.7			0.8			0.9		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
MEAN	0.0830	0.0689	75.50%	0.0829	0.0689	75.54%	0.0829	0.0689	75.54%	0.0830	0.0689	75.54%
LAST	0.0824	0.0608	60.59%	0.0865	0.0664	66.36%	0.0910	0.0730	72.99%	0.0962	0.0812	81.16%
GAIN	0.0401	0.0292	31.89%	0.0577	0.0447	48.96%	0.0911	0.0684	74.71%	0.1103	0.0784	85.78%
E^2GAN	0.0570	0.0401	43.77%	0.0630	0.0449	48.83%	0.0682	0.0507	55.27%	0.0795	0.0598	65.17%
SolarGAN	0.0334	0.0232	26.07%	0.0378	0.0254	28.09%	0.0418	0.0304	35.50%	0.0496	0.0363	40.35%
Transformer	0.0374	0.0295	32.53%	0.0388	0.0303	33.52%	0.0413	0.0326	34.37%	0.0424	0.0334	36.87%
TimesNet	0.0337	0.0269	29.66%	0.0451	0.0365	40.35%	0.0500	0.0383	42.25%	0.0512	0.0399	44.03%
iTransformer	0.0323	0.0229	25.50%	0.0354	0.0235	27.46%	0.0385	0.0260	30.15%	0.0434	0.0281	32.58%
DINCAE	0.0316	0.0240	26.38%	0.0333	0.0264	29.78%	0.0337	0.0283	32.19%	0.0369	0.0307	34.93%
S2-STI	**0.0306**	**0.0228**	**25.10%**	**0.0309**	**0.0232**	**25.60%**	**0.0332**	**0.0233**	**26.04%**	**0.0350**	**0.0241**	**26.45%**
Δ	3.16%↑	0.44%↑	1.57%↑	7.21%↑	1.28%↑	6.77%↑	1.48%↑	10.38%↑	13.63%↑	5.15%↑	14.23%↑	18.82%↑

improvement of S2-STI on these evaluation metrics is also obvious and reaches more than 13%, 14%, and 8%, respectively. Among the traditional imputation methods, MAEN performs the worst as it does not consider spatiotemporal correlations in Chl-a data. The LAST method, which considers local temporal correlations, outperforms MEAN at low missing rates, but performs worse than MEAN at high missing rates due to unavailability of recent historical observations. Among the other deep learning methods, E^2GAN and GAIN perform the worst as they struggle to learn the temporal correlations in highly sparse satellite data. Transformer performs moderately well and achieves quite good performance at high missing rates due to its nonautoregressive structure which avoids error accumulation. TimesNet and SolarGAN perform slightly worse than S2-STI because they do not consider the spatial features of the Chl-a data. Since iTransformer and DINCAE can effectively capture the spatial features of Chl-a data, they also achieve quite good performance.

According to Tables I and II, with the increase in missing rate, the performance of all the imputation methods decreases due to the reduction of available Chl-a observation data for model training and data imputation. With a small missing rate, e.g., less than 0.5, S2-STI generally outperforms the strong baselines, i.e., TimesNet and SolarGAN by a small margin. However, with further increase in missing rate, the performance gap between S2-STI and these two baseline methods gradually becomes more pronounced, indicating that S2-STI is good at data imputation for highly sparse Chl-a data.

Additionally, Fig. 14 illustrates the performance lines of S2-STI and baseline methods on the two datasets while varying the data missing rate. Obviously, with the increase in data missing rate, S2-STI exhibits moderate performance decline and presents good performance even when the missing rate reaches 0.9.

D. Ablation Studies

This section conducts ablation studies to evaluate the rationality of the structural design of S2-STI and the validity of the exogenous data.

1) Ablation Study on the Architecture Design of S2-STI: To evaluate the effectiveness of those proposed components in S2-STI, we design different variants of S2-STI by removing certain components, i.e.,

1) *w/o T*: removing the temporal embedding;
2) *w/o S*: removing the spatial embedding;

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TABLE II

RESULTS OF S2-STI AND BASELINE DATA IMPUTATION METHODS ON THE CHA-YRE DATASET WHILE VARYING THE MISSING RATE, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND ↑ REPRESENTS THE MINIMUM INCREASE RATIO

Method	0.2			0.3			0.4			0.5		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
MEAN	5.2936	3.5425	84.90%	5.2988	3.5453	84.91%	5.3019	3.5440	84.90%	5.3048	3.5441	84.90%
LAST	4.4840	2.7760	66.53%	4.5789	2.8733	68.81%	4.6913	2.9833	71.47%	4.8220	3.1117	74.55%
GAIN	3.6402	2.1284	52.79%	3.7023	2.2986	56.60%	4.0177	2.4067	59.22%	4.0610	2.4811	61.10%
E^{2}GAN	3.1056	1.8616	44.19%	3.2216	1.8758	45.26%	3.4437	2.0056	49.00%	3.7113	2.2197	53.15%
SolarGAN	2.9842	1.8191	45.18%	3.0008	1.8639	46.25%	3.0286	1.8736	46.52%	3.0471	1.8895	46.92%
Transformer	3.0561	2.1433	51.31%	3.2011	2.1453	51.33%	3.2104	2.2586	54.06%	3.2480	2.2918	55.07%
TimesNet	3.0995	2.0020	47.94%	3.1117	2.0291	48.55%	3.1437	2.0307	48.60%	3.1501	2.0411	49.42%
iTransformer	3.0627	1.9091	**41.44%**	3.1755	1.9879	43.19%	3.1825	1.9942	43.37%	3.3080	2.0824	45.32%
DINCAE	3.7012	1.8957	41.51%	3.7361	1.9325	45.62%	3.7854	1.9748	46.12%	3.8484	2.0174	47.60%
S2-STI	**2.8360**	**1.7428**	42.81%	**2.8633**	**1.7593**	**43.02%**	**2.8744**	**1.7638**	**43.19%**	**2.8837**	**1.7674**	**43.34%**
Δ	4.97%↑	4.19%↑	3.31%↓	4.58%↑	5.61%↑	0.39%↑	5.09%↑	5.86%↑	0.42%↑	5.36%↑	6.46%↑	4.37%↑
Method	0.6			0.7			0.8			0.9		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
MEAN	5.3088	3.5451	84.91%	5.3058	3.5438	84.90%	5.3077	3.5435	84.90%	5.3078	3.5438	84.90%
LAST	4.9653	3.2602	78.08%	5.1215	3.4303	82.18%	5.3031	3.6320	87.02%	5.5175	3.8772	92.89%
GAIN	4.2544	2.4982	61.49%	4.4730	2.5326	62.45%	5.2569	3.0263	74.59%	5.3899	3.1420	87.34%
E^{2}GAN	3.9076	2.4022	58.32%	4.1893	2.5648	70.69%	4.3397	3.0290	74.97%	4.5481	3.3978	79.33%
SolarGAN	3.1049	1.9474	48.24%	3.2291	2.0520	50.86%	3.3773	2.1914	52.24%	3.4453	2.3088	54.57%
Transformer	3.2888	2.3520	56.28%	3.4008	2.3538	56.37%	3.5084	2.4159	58.43%	3.9479	2.7309	65.36%
TimesNet	3.2444	2.1379	51.16%	3.2826	2.1400	51.22%	3.3655	2.1549	51.58%	3.5238	2.3165	55.49%
iTransformer	3.3568	2.1145	46.01%	3.4783	2.2030	47.91%	3.4992	2.4654	48.19%	3.5988	2.2884	49.80%
DINCAE	3.8697	2.0634	48.84%	3.8822	2.0984	49.09%	3.9283	2.1050	49.54%	4.0920	2.3600	55.26%
S2-STI	**2.8942**	**1.7826**	**43.51%**	**2.8969**	**1.7891**	**43.81%**	**2.9065**	**1.8026**	**44.29%**	**2.9851**	**1.8376**	**45.36%**
Δ	6.79%↑	8.46%↑	5.43%↑	10.29%↑	12.81%↑	8.56%↑	13.64%↑	14.37%↑	8.09%↑	13.36%↑	19.70%↑	8.92%↑

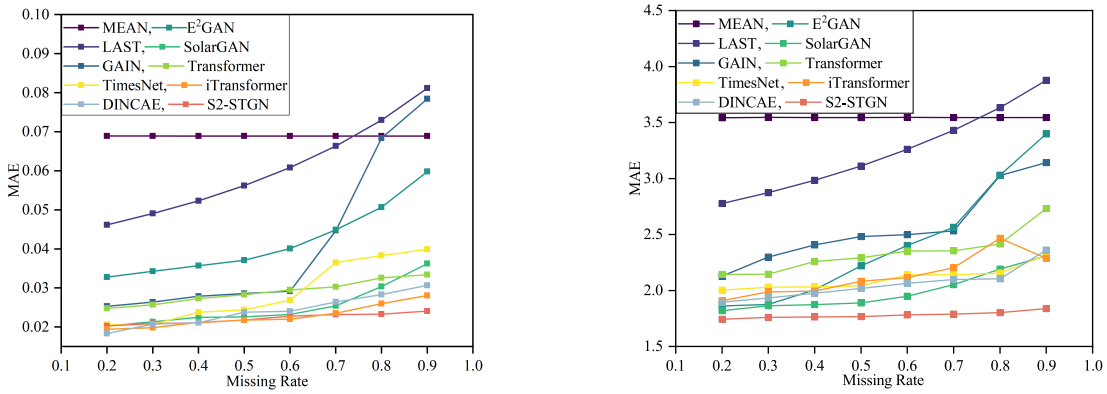


Fig. 14. MAE of CHA-EAST and CHA-YRE datasets with different missing rates. (a) MAE of CHA-EAST. (b) MAE of CHA-YRE.

3) *w/o B*: removing the bidirectional GRU;
4) *w/o* $\mathcal{L}_{imput}$: removing the imputation loss $\mathcal{L}_{\text{imput}}$.

Table III shows the results of these variants and the original S2-STI, where the data missing rate is also set to 0.5. Obviously, after removing either of the four components, the performance of S2-STI decreases, which indicates the effectiveness and necessity of these components. The temporal embedding and spatial embedding can effectively learn the spatiotemporal features of exogenous data, and the bidirectional GRU can well learn the distribution of Chl-a data in the generator. In addition, the imputation loss $\mathcal{L}_{\text{imput}}$ plays a crucial role during the generator training based on observed values.

2) Ablation Study on the Impacts of Exogenous Data: To explore the impacts of exogenous data, we evaluate the variants of S2-STI as follows:

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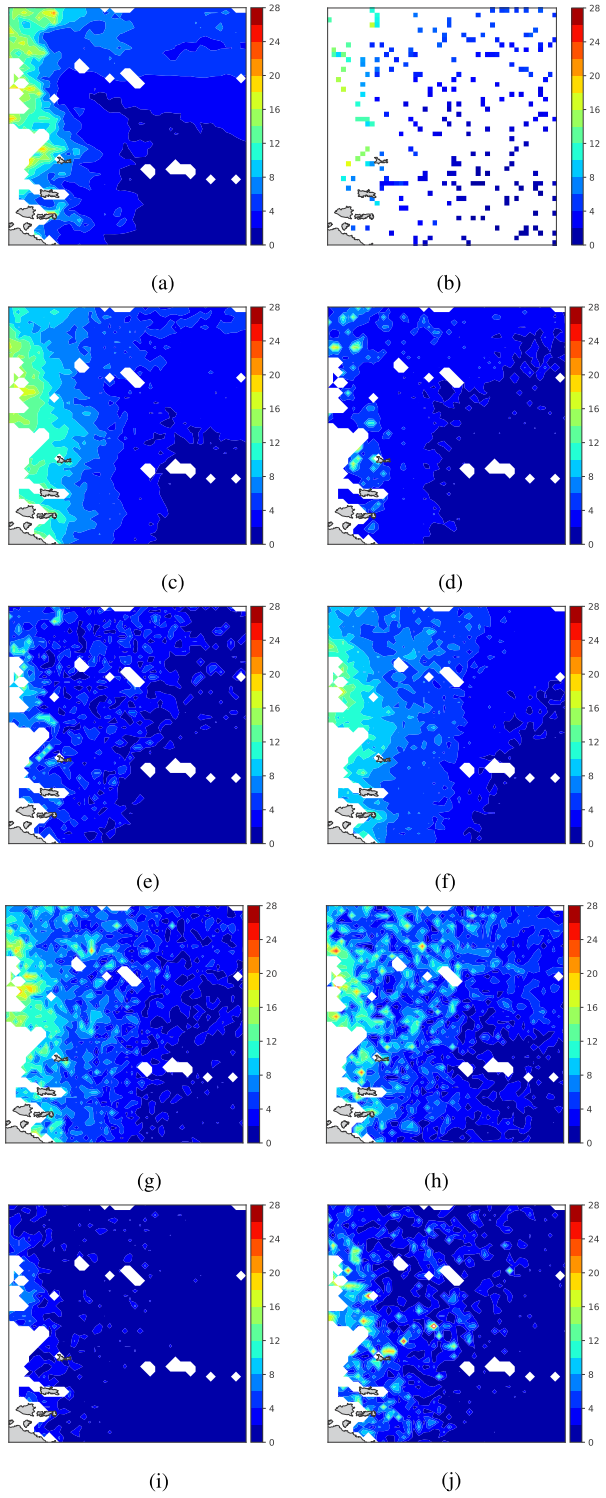


Fig. 15. Comparison on February 10, 2020. (a) Raw data. (b) Mask-raw. (c) S2-STI. (d) DINCAE. (e) iTransformer. (f) SolarGAN. (g) TimesNet. (h) Transformer. (i) E^{2}GAN. (j) GAIN.

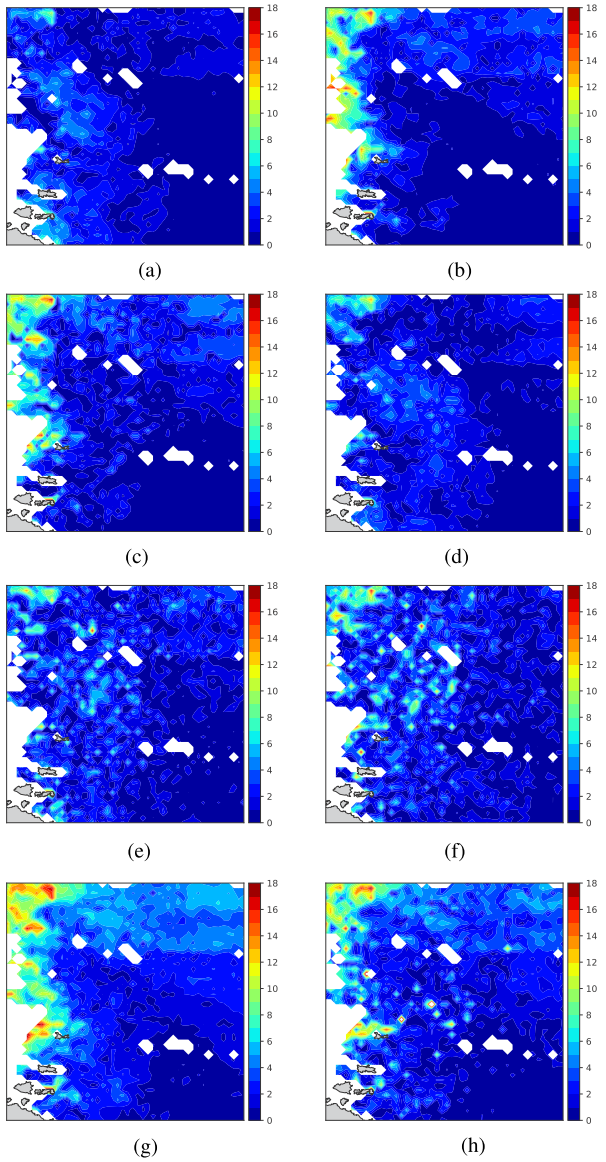


Fig. 16. MAE on February 10, 2020. (a) S2-STI, $MAE_{avg} = 1.4856$. (b) DINCAE, $MAE_{avg} = 2.1167$. (c) iTransformer, $MAE_{avg} = 1.5632$. (d) SolarGAN, $MAE_{avg} = 1.6213$. (e) TimesNet, $MAE_{avg} = 1.9056$. (f) Transformer, $MAE_{avg} = 2.1409$. (g) E^{2}GAN, $MAE_{avg} = 3.4415$. (h) GAIN, $MAE_{avg} = 2.7000$.

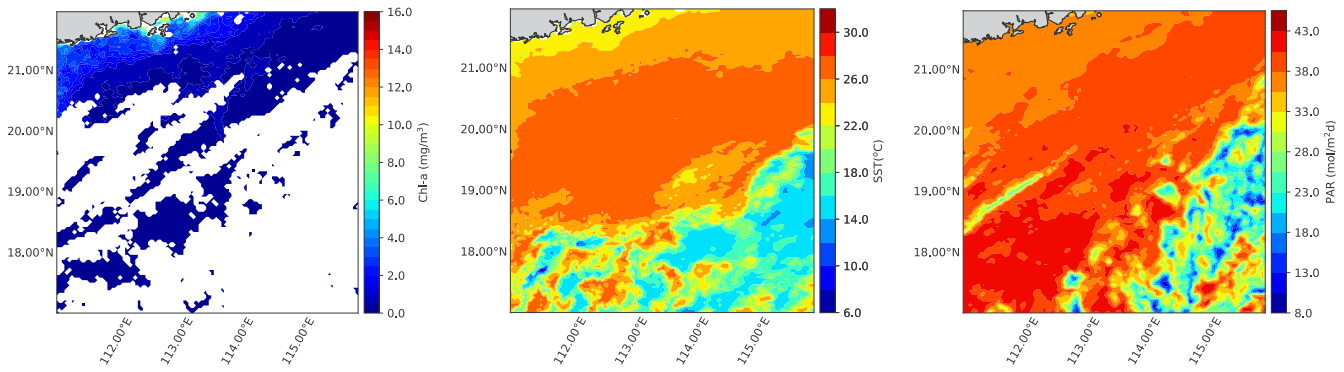
1) *w/o PAR*: removing the exogenous data PAR;
2) *w/o SST*: removing the exogenous data SST;
3) *w/o PAR & SST*: removing the exogenous data SST and PAR.

According to the results in Table IV, both PAR and SST help to improve the imputation accuracy of Chl-a data. It can be seen from the results that SST data contributes more to Chl-a data imputation than PAR, which is consistent with the correlation demonstrated in Fig. 3.

TABLE III
RESULTS OF S2-STI AND ITS VARIANTS AFTER REMOVING CERTAIN COMPONENTS

Model	CHA-EAST		CHA-YRE	
	RMSE	MAE	RMSE	MAE
w/o T	0.0313	0.0227	2.8980	1.7900
w/o S	0.0323	0.0227	2.9049	1.8031
w/o B	0.0324	0.0230	2.9048	1.8214
w/o $\mathcal{L}_{imput}$	0.0315	0.0230	2.9358	1.8525
S2-STI	**0.0296**	**0.0218**	**2.8837**	**1.7674**

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(a) (b) (c)

Fig. 17. Spatial distributions of Chl-a, SST, and PAR in CHA-SOUTH dataset on November 27, 2023, where the white areas have no data, and the gray areas are land. (a) OC-CCI Chl-a. (b) AVHRR SST. (c) MODIS-Terra PAR.

TABLE IV
RESULTS OF S2-STIAND ITS VARIANTS AFTER REMOVING CERTAIN EXOGENOUS DATA

Model	CHA-EAST		CHA-YRE	
	RMSE	**MAE**	**RMSE**	**MAE**
w/o PAR	0.0314	0.0225	2.9055	1.7978
w/o SST	0.0315	0.0228	2.9622	1.8545
w/o PAR & SST	0.0326	0.0262	3.1771	2.2876
S2-STI	**0.0296**	**0.0218**	**2.8783**	**1.7674**

V. DISCUSSION

A. Superiority of S2-STIfor Highly Sparse Data

S2-STI model is tailored for filling missing values in highly sparse Chl-a data by fusing multisource satellite data. To clearly demonstrate the advantages of S2-STI, we carry out the discussion on its excellent ability in learning the data distribution during the data imputation process on the highly sparse CHA-YRE dataset. To this end, we choose a representative data sample with almost no missing values from February 10, 2020, thus providing enough true values to evaluate the accuracy of the filled values by data imputation methods. Fig. 15 illustrates the raw data, the masked data with a mask ratio of 0.9, and the generated data of S2-STI and seven baseline methods, i.e., DINCAE, iTransformer, SolarGAN, TimesNet, Transformer, E^{2}GAN, and GAIN, on three data samples.

The visualization results show that the data generated by the E^{2}GAN and GAIN deviates significantly from the raw data, especially near the coastline, indicating their poor ability in learning the distribution of Chl-a data. Transformer and DINCAE are able to roughly maintain the shape of the data distribution, but the data they generate still has some obvious differences from the raw data. The data generated by TimesNet and SolarGAN generally follows the distribution of the original data, but some data points still show significant deviations from the original values. iTransformer, by generating embeddings for each location, achieves better imputation results than Transformer. In contrast, the data distribution learned by S2-STI is more in line with the distribution of the raw Chl-a data, which indicates that the S2-STI has a robust ability to learn the distribution of highly sparse Chl-a data.

Fig. 16 visualizes the MAE and the corresponding average MAE value (i.e., MAE$_{avg}$) of S2-STI and the seven baseline methods on the selected data sample. S2-STI exhibits a low MAE, especially for the coastal areas where the Chl-a data has more complex spatiotemporal dependencies and dynamics.

B. Applicability of S2-STI

We evaluate S2-STI in a larger region of size 5 × 5, i.e., the Northern South China Sea (referred to as CHA-SOUTH) [111.00°E–116.00°E, 17.00°N–22.00°N], to demonstrate that the S2-STI is also applicable to other ocean regions. Fig. 17 illustrates the spatial distributions of Chl-a, SST, and PAR in CHA-SOUTH dataset on one selected day. Table V shows the performance comparison between the S2-STI and the representative deep learning-based data imputation methods at missing rates of 0.2, 0.5, and 0.9. We can see from Table V that S2-STI obviously outperforms the baseline methods when the data missing rate is above 0.5. Note that the results of iTransformer and GAIN are not reported here. iTransformer calculates the spatial-centered attention, thus requiring significant storage and making it difficult for us to run iTransformer on the CHA-SOUTH dataset. As for GAIN, it is a GAN-based method and fails on the CHA-SOUTH dataset due to loss explosion.

In addition, according to the experimental evaluation, there exists obvious performance difference for S2-STI and the baseline methods when applied on the CHA-EAST and CHA-YRE datasets. Specifically, CHA-YRE corresponds to the region around the Yangtze River Estuary and the Chl-a concentration is of high dynamics and uncertainty due to the influence from human, land, river, etc. In contrast, CHA-EAST is an offshore region with more stable water conditions and Chl-a concentration than CHA-YRE. Therefore, the sparse Chl-a data imputation performance of S2-STI and the baseline methods on CHA-EAST is generally better than that on CHA-YRE. According to the results in Table II, S2-STI is still superior to the existing Chl-a data imputation methods in regions like CHA-YRE of high dynamics and uncertainty since

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TABLE V

RESULTS OF S2-STIAND BASELINE DATA IMPUTATION METHODS ON THE CHA-SOUTH DATASET WHILE VARYING THE MISSING RATE, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND ↑ REPRESENTS THE MINIMUM INCREASE RATIO, ↓ REPRESENTS THE MAXIMUM DECREASE RATIO

Method	0.2			0.5			0.9		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
GAIN	0.0345	0.0253	27.64%	0.0364	0.0264	28.88%	0.0369	0.0279	30.50%
E^2GAN	0.7283	0.2196	61.04%	0.7655	0.2380	66.13%	0.9976	0.3156	87.59%
SolarGAN	0.5921	0.1838	51.20%	0.6068	0.1925	53.65%	0.6715	0.2184	60.78%
Transformer	0.6087	0.2042	54.22%	0.6721	0.2367	62.80%	0.7419	0.2606	69.16%
TimesNet	**0.5797**	0.1877	**49.82%**	0.6200	0.2079	55.16%	0.6644	0.2294	60.87%
DINCAE	0.7702	0.1954	52.71%	0.8246	0.2035	53.34%	0.8537	0.2173	56.32%
S2-STI	0.5831	**0.1793**	49.85%	**0.6139**	**0.1890**	**52.56%**	**0.6185**	**0.1906**	**52.97%**
Δ	0.59%↓	4.48%↑	0.06%↓	0.98%↑	7.13%↑	1.46%↑	6.91%↑	12.29%↑	5.95%↑

TABLE VI

NUMBER OF PARAMETERS (IN MILLION) AND THE RUNNING TIME OF EACH TRAINING EPOCH (IN SECONDS) FOR S2-STI AND THE BASELINE METHODS ON CHA-YRE DATASET

Model	Parameters (M)	Time Cost (seconds/epoch)
GAIN	10.68	22.10
E^2GAN	0.91	9.5
SolarGAN	2.93	4.6
Transformer	13.88	3.0
TimesNet	0.97	3.0
iTransformer	0.22	7.5
DINCAE	0.17	30.0
S2-STI	26.24	15.5

it can effectively learn the spatiotemporal patterns of Chl-a and take full advantages of exogenous data to reduce the uncertainty. Therefore, S2-STI has high generalization ability and is applicable to various water types in ocean.

C. Computational Efficiency

Table VI presents the number of parameters (in millions) and the running time of each training epoch (in seconds) for S2-STI and the baseline methods on CHA-YRE dataset. S2-STI improves the imputation performance of highly sparse Chl-a data by incorporating exogenous datasets (i.e., SST and PAR). This introduces more parameters for learning the spatiotemporal representations of exogenous data during the training process. However, the running time of S2-STI is still comparable to that of baseline methods. Considering the excellent imputation performance of S2-STI, the time cost and parameter count of S2-STI are reasonable and acceptable.

VI. CONCLUSION

In this work, we proposed a novel self-supervised model termed S2-STI for highly sparse Chl-a data imputation by fusing multisource satellite data, i.e., Chl-a, SST, and PAR. S2-STI first learns the high-quality spatiotemporal representations of SST and PAR via a STGN, and fuses them with the highly sparse Chl-a data to obtain comprehensive information about the missing values in Chl-a data. Then, S2-STI learns the Chl-a data distribution based on the obtained comprehensive information via a self-supervised training strategy and uses the learned distribution to generate complete Chl-a data for achieving the final data imputation. The experiments on two real Chl-a datasets demonstrate that the performance of S2-STI is significantly better than the baseline imputation methods, and has the most modest performance decline as the missing rate increases.

As for future work, the time complexity of S2-STI and the existing deep learning-based Chl-a data imputation methods is still high, making it difficult to conduct Chl-a data imputation over the global scale. To address this issue, we plan to further study the lightweight version of S2-STI to reduce the time complexity while keeping the data imputation accuracy. In addition, it is promising to explore the collaborative data imputation for multiple ocean factors, e.g., Chl-a, SST, and PAR, and take full advantage of their correlations to enrich the information and knowledge for data imputation. Moreover, combining data-driven methods, including both machine learning methods and deep learning methods, and statistical methods for Chl-a data imputation to achieve both high interpretability and accuracy is also an interesting direction.

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Shuyu Wang received the master's degree in computer science and technology from China University of Petroleum, Dongying, Shandong, China, in 2021. She is currently pursuing the Ph.D. degree with the Department of Computer Science and Technology, Tongji University, Shanghai, China.

Her research interests include remote sensing data processing, spatial–temporal data imputation, and deep learning.

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Wengen Li (Member, IEEE) received the B.Eng. and Ph.D. degrees in computer science from Tongji University, Shanghai, China, in 2011 and 2017, respectively, and the dual Ph.D. degree in computer science from Hong Kong Polytechnic University, Hong Kong, in 2018.

He is currently an Associate Professor at the Department of Computer Science and Technology, Tongji University. His research interests include multimodal data analytics, and spatiotemporal data analytics for urban computing and ocean computing.

Dr. Li is a member of China Computer Federation (CCF) and a member of Association for Computing Machinery (ACM).



Jihong Guan received the bachelor's degree from Huazhong Normal University, Wuhan, China, in 1991, the master's degree from Wuhan Technical University of Surveying and Mapping (merged into Wuhan University since 2000), Wuhan, in 1998, and the Ph.D. degree from Wuhan University, Wuhan, in 2002.

She is currently a Professor at the Department of Computer Science and Technology, Tongji University, Shanghai, China. Before joining Tongji University, she served at the Department of Computer, Wuhan Technical University of Surveying and Mapping, from 1991 to 1997, as an Assistant Professor, and an Associate Professor, since August 2000, respectively. She was an Associate Professor from 2000 to 2003 and a Professor since 2003 at the School of Computer, Wuhan University. Her research interests include databases, data mining, distributed computing, bioinformatics, and geographic information systems (GIS).



Xiwei Liu (Senior Member, IEEE) was born in 1982, Shandong, China. He received the Ph.D. degree in applied mathematics from Fudan University, Shanghai, China, in 2008.

He was a Postdoctoral Researcher with the Department of Physics, Fudan University, from 2008 to 2010. Then, he joined the Department of Computer Science and Technology, Tongji University, Shanghai. He has been a Professor with the Department of Computer Science and Technology, Tongji University, since December 2016. His research interests include nonlinear dynamical systems, complex networks, and neural networks.

Dr. Liu received the Natural Science Award from the Ministry of Education of China in 2015.



Yichao Zhang (Member, IEEE) received the Ph.D. degree in computer science and technology from Tongji University, Shanghai, China, in 2012.

Currently, he is an Associate Professor at the Department of Computer Science and Technology, Tongji University. His research interests include information diffusion, link prediction, modeling of weighted networks, random diffusion on weighted networks, and evolutionary games on networks.



Shuigeng Zhou (Senior Member, IEEE) received the bachelor's degree from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 1988, the master's degree from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1991, and the Ph.D. degree in computer science from Fudan University, Shanghai, China, in 2000.

He served at Shanghai Academy of Spaceflight Technology, Shanghai, as an Engineer, from 1991 to 1997, and a Senior Engineer, since August 1995, respectively. He was a Postdoctoral Researcher at the State Key Lab of Software Engineering, Wuhan University, Wuhan, from 2000 to 2002. He is currently a Professor at the School of Computer Science, Fudan University. He has published more than 200 papers in domestic and international journals (including ACM TITS, IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS (TPDS), *Very Large Data Bases (VLDB) Journal*, IEEE/ACM TRANSACTIONS ON COMPUTATIONAL BIOLOGY AND BIOINFORMATICS (TCBB), and *Bioinformatics*) and conferences [including Special Interest Group on Management of Data (SIGMOD), VLDB, IEEE International Conference on Data Engineering (ICDE), ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (SIGKDD), ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), Association for the Advancement of Artificial Intelligence (AAAI), International Joint Conference on Artificial Intelligence (IJCAI), IEEE International Conference on Computer Vision (ICCV), IEEE Conference on Computer Vision and Pattern Recognition (CVPR), ACM-SIAM Symposium on Discrete Algorithms (SODA), Intelligent Systems for Molecular Biology (ISMB), and Research in Computational Molecular Biology Conference (RECOMB)]. His research interests include big data management and analysis, artificial intelligence, and bioinformatics.

Dr. Zhou is a member of Association for Computing Machinery (ACM).

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